Multi Sensor Data Fusion Based Expert System in Condition Monitoring and Fault Detection: An Induction Motor Case Study

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Abstract— **Condition monitoring is a predictive maintenance procedure, which is used to provide warning and predict faults of a system at an early stage, so catastrophic breakdown of the system can be avoided. An efficient condition monitoring method can reduce the cost of breakdown maintenance, increase the working life of system, and reduce the spare part inventory. Condition monitoring system acquires raw data of different system parameters using different sensors. Different signal processing techniques are implemented on the raw data to extract vital clues and estimate the probable faults in the machine. This paper gives an overview of condition monitoring system of induction motor and gives an overview of implementation of multi sensor data fusion technique to improve the efficiency of condition monitoring system.**

Keywords- Condition monitoring, multi sensor data fusion, signal processing.

I. INTRODUCTION

Electric motors account for 95% of all prime movers in industrialized nations, and among these, three-phase induction motors consume typically 40% to 50% of all generated electrical energy. Induction motors are inherently reliable and require minimum maintenance.

Condition monitoring of any mechanical, electro mechanical or electro static devices is a challenging task. There are different methods of condition monitoring of an induction motor including monitoring of vibration, acoustic emission, and motor current signature analysis etc. These monitoring techniques except motor current signature analysis technique require high precision sensors, DAQ cards and sophisticated signal processing analysis procedures.

An efficient condition-monitoring scheme is one that provides warning and predicts the faults at early stages. Condition monitoring system acquires raw data of different system parameters using different sensors. Different signal processing techniques are implemented on the raw data to extract vital clues and estimate the probable faults in the machine. The problem with this approach is that the results require constant human interpretation. To automate the diagnostic process, recently a number of soft computing techniques have been proposed [5, 6, 7, 8]. The use of soft computing techniques increases the precision and accuracy of the monitoring systems [9].

A single electric motor can have multiple faults, which can't be detected using a single sensor. Multiple sensors are used to acquire multiple signals and multi sensor fusion technique is implemented to detect the faults of a motor.

A Multi Sensor Fusion & Condition Monitoring

Data fusion techniques combine data from multiple sensors to achieve improved accuracies and more specific inferences than could be achieved by the use of a single sensor alone. Multi sensor data fusion is an emerging technology applied to different areas such as automated target recognition, battlefield surveillance, and guidance and control of autonomous vehicles, monitoring of complex machinery, medical diagnosis, and smart buildings. Techniques for multi sensor data fusion are drawn from a wide range of areas including artificial intelligence, pattern recognition, statistical estimation etc.

Data collected from different sensors can be fused in different levels. The data fusion takes place in the following levels.

1. Data level (If the data are commensurate)
2. State vector level
3. Feature level
4. Decision level

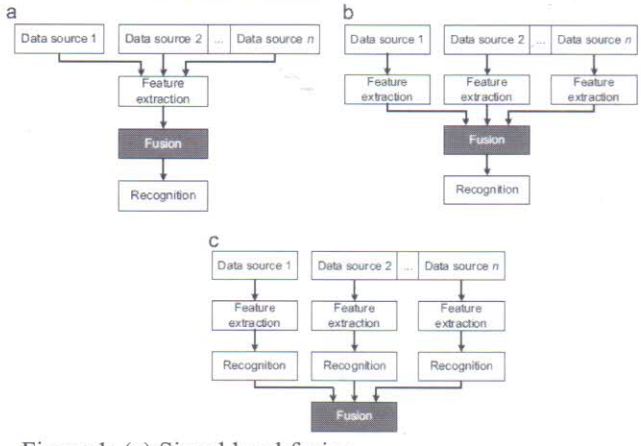


Figure 1: (a) Signal level fusion,
(b) feature level fusion,
(c) decision level fusion

Figure 1 (a), (b) and (c) shows the signal level fusion, feature level fusion and decision level fusion.

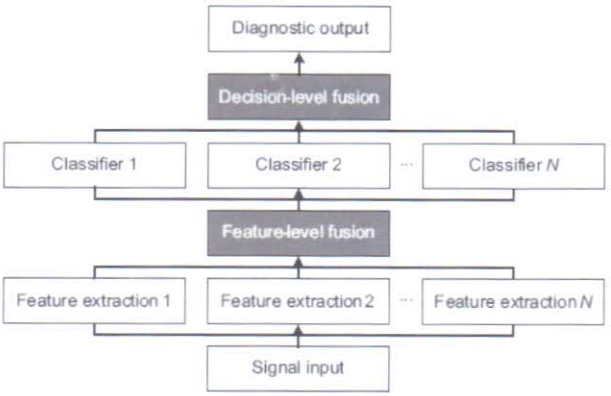


Figure 2: Hybrid architecture

Figure 2 shows the hybrid architecture, where feature level and decision level fusion is implemented to make a diagnosis.

B Condition Monitoring of Induction Motor

Induction motors are most widely used electro-mechanical machines in industry. The wide use of induction motor is because of their low cost, reasonably small size, ruggedness, low maintenance. Although these motors are very reliable, these are subjected to different modes of faults or failures. These faults can be inherent to the machine itself or due to operating conditions. Figure 3 shows the different faults in induction machine [].

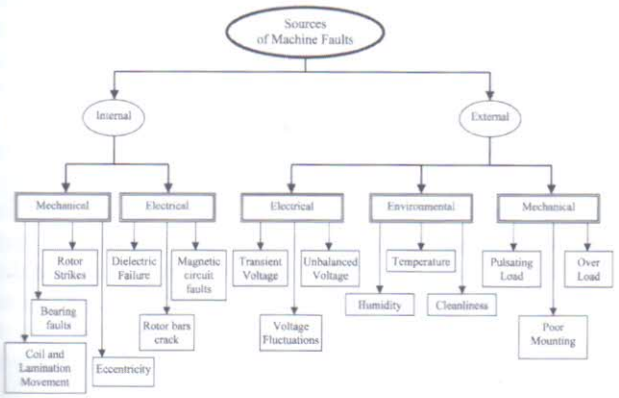


Figure 3: Faults in induction machine

Basically the faults occurring in induction machine are divided in to two parts, i.e faults due to internal factors and faults due to external factors. Some of the internal faults are eccentricity fault, broken rotor bar fault, bearing fault, dielectric failures etc. Faults can occur in induction machines due to some external factors like high temperature, high humidity, overload, unbalanced voltage, transient voltage, voltage fluctuations etc.

According to IEEE and EPRI (Electric Power Research Institute) survey results, the faults and losses of induction motor are summarized below.

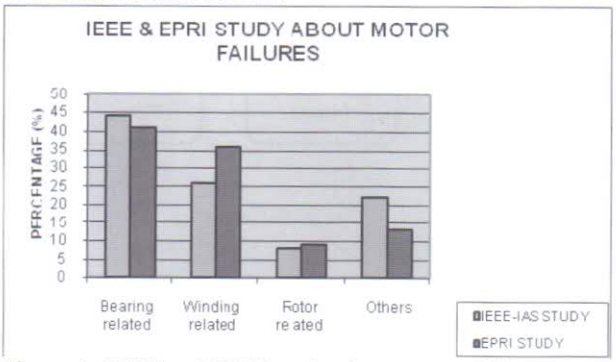


Figure 4: IEEE and EPRI study about motor failures

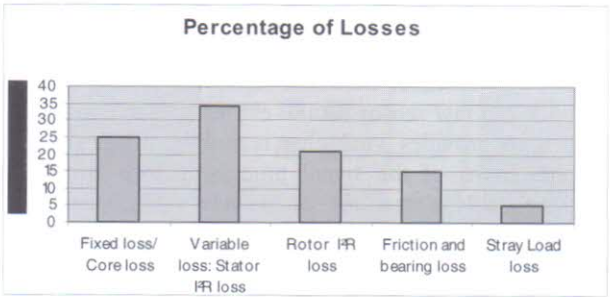


Figure 5: Percentage of losses in induction motor

Traditionally maintenance procedures in industry follow two approaches.

(i) Preventive Maintenance

(ii) Breakdown Maintenance.

In preventive maintenance, maintenance operation is carried out in a regular interval of time. In breakdown maintenance, maintenance is carried out when the breakdown occurs. However, now a day with implementation of different scientific approach, predictive maintenance is possible. In predictive maintenance, different signal processing approaches are used to detect the early signs of defect and suitable actions are taken before any kind of catastrophic failure occurs.

Different fault diagnosis methods are used to detect different faults of induction motor. These techniques are summarized below.

There are basically three techniques for predictive maintenance namely, model based technique, signal processing techniques and soft computing technique.

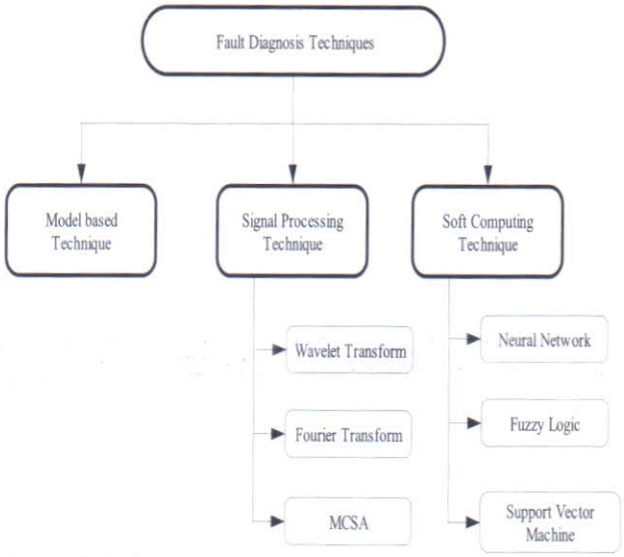


Figure 6: Induction motor fault diagnosis technique

Signal processing technique is widely used to detect faults in induction machine. Different signal processing algorithms are applied in raw sensor data to extract valuable information, which then provides vital clue regarding the health of the machine. Some of the signal processing techniques widely used for fault detections is wavelet transform, Fourier transform, Short Time Fourier Transform, Motor Current Signature Analysis, Power Spectral Density etc. Soft computing techniques like neural network and fuzzy logic can also be used for fault detection in induction motor. Different supervised and unsupervised classification techniques like Support Vector Machines and Self Organizing Maps respectively can be implemented in raw sensor data to ascertain the health of the machine.

II. Related Works

Lane M D Owsley et.al (1997), uses SOFM to extract feature from vibration signal of induction motor and uses HMM to classify different fault data [5]. Zhongming et al (2001), used WPD to detect both air gap eccentricity and broken rotor bar after giving a brief detail about the wavelet and feature extraction [7]. G K Singh (2002), reviewed the research aspect of multi phase induction machine drives. Multi phase induction machine is a potential alternative of the conventional 3 phase induction machine. This paper reviews the current development and issues and future challenges to design a multi phase induction machine drive [9]. G K Singh et.al (2003) studied the use of signal processing techniques to extract valuable information about machine health condition. The author studied about online and offline condition monitoring techniques [11]. G K Singh et.al (2003), carried out a survey of different techniques used in induction motor condition monitoring and fault diagnosis [12]. B Samanata et.al (2003), compared the working of ANN and SVM for fault detection and condition monitoring of induction motor. In case of SVM, features are optimized with the help of GA [13]. Romero-Troncoso Rene de Jesus et.al (2004) had a study about FPGA based online tool breakage detection system for CNC milling machine. It used DWT and autocorrelation algorithm to detect tool breakage [14]. Hamidi et.al (2004), presented the detection of mixed eccentricity fault using WPD, which was done by a modified winding function [15]. Antonino et al (2005), presented the detection and diagnosis of mixed eccentricities and rotor asymmetries with different sizes and conditions as well as effective oscillations due to the load torque or voltage that were studied individually [19]. Antonino et al (2006), presented many cases for fault diagnosis (mixed eccentricity, broken rotor, inter-turn and inter-coil stator short-circuits) using DWT at start up current of induction machines according to the stator parallel branches [20]. Fulufhelo V Nelwamondo et.al (2006) has reviewed feature extraction techniques to extract both linear and non linear features from vibration signal of a rotating machine. Features are extracted by MFD while considering two parameters like MFCC and Krutosis and bearing fault is classified using GMM and HMM. The result indicates that though computationally expensive HMM is a better classifier than GMM, while classifying bearing faults from time domain vibration signal [23]. Tian Han et.al (2006), has used wavelet transform, ANN and GA to predict the faults in induction machine. These techniques were implemented in stator current and the fault of induction machine is predicted. In pre processing of stator current signal, wavelet transform is used. GA is used to select the most significant feature vector, while ANN is used to predict the fault. The combination of all these techniques improves the diagnosis accuracy [24]. Abhinav Saxena and Ashraf Saad (2007), uses GA to select smaller subset of features that together form a genetically fit family for successful fault identification and classification task [30]. Jafar and Javad (2007) used Meyer wavelet in the WP structure to detect the bearing defect using SCA with energy comparison as the fault index [32]. Bo Suk Yang et.al (2007), surveys the use of support vector machine (SVM) for machine condition monitoring and fault detection. SVM produces higher accuracy in classification tasks for machine condition monitoring and fault diagnosis. Bin and Paghda (2008) used the wavelet to detect the broken rotor bar, eccentricity and bearing due to current, voltage and instantaneous power, while the signals to noise ratio of the spectral components were examined under varying load condition of the single phase in the active cycle [33]. Jawad and Mansour (2009) presented a review for most important indexes in the different types of eccentricities faults in the induction motors as well as the consequences and effects [38]. Yixiang et al (2010), used the lean model to assess the machine performance via DWT for the vibration and bearing induction motor faults [41]. C. Rodriguez-Donate et.al implemented an embedded system with FPGA and SOC (System on Chip) for online monitoring of induction motor failures by measuring the vibration transient signals at the start-up with discrete wavelet transform. Franco-Gasca et.al (2009), presents a hardware signal processing unit implemented in a single field-programmable gate array (FPGA) for acquisition, conditioning, and basic

signal monitoring in several machining processes. This model is reconfigurable and scalable so that it may be adapted to diverse conditions as an economical standalone unit since it does not require either computers or microprocessors. Luis Miguel Contreras-Medina et.al (2010), developed a low-cost FPGA based on a multichannel vibration analyzer; this is capable of providing an automatic diagnosis of the motor state carrying out online continuous monitoring [39].

III. HARDWARE IMPLEMENTATION

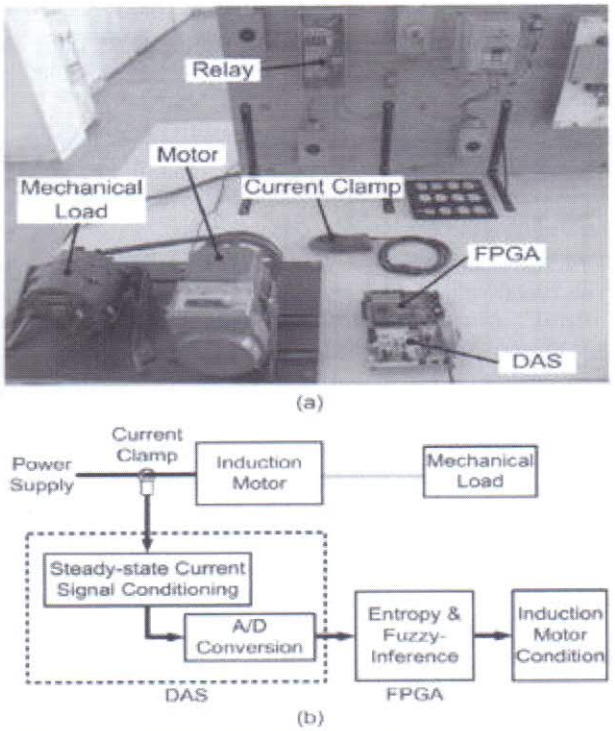


Figure 7: Condition monitoring system with FPGA

Figure 7 shows the condition monitoring of induction motor system with FPGA [42]. Figure 8 shows the FPGA board indicating the fault. Figure 9 and figure 10 shows the PC based condition monitoring kit.

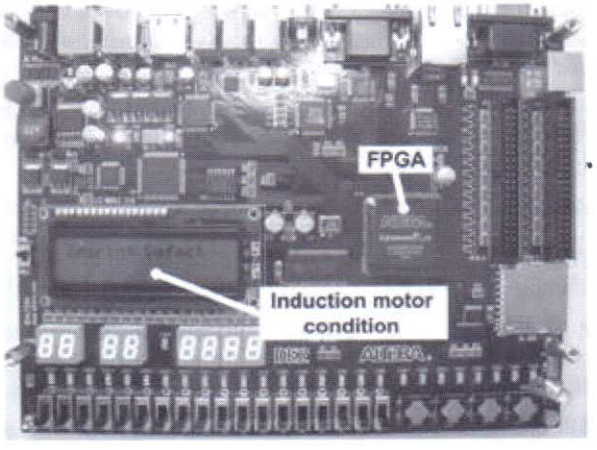


Figure 8: FPGA board showing the condition of induction motor

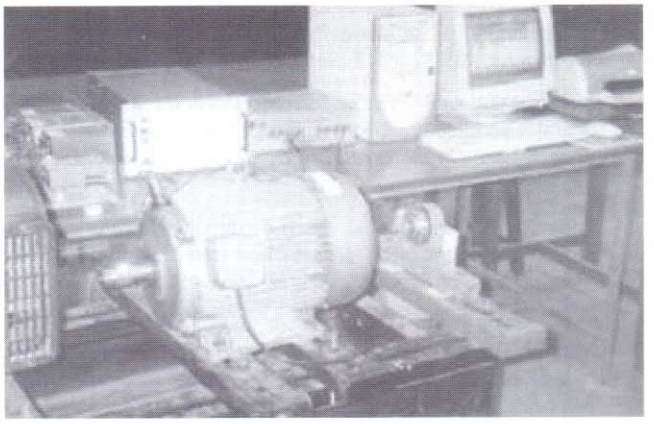


Figure 9: Condition monitoring for 7.5 KW, 415 V, 1440 rpm induction motor



Figure 10: Condition monitoring kit for induction motor

This section gives an overview of hardware implementation of condition monitoring system. Figures 7-10 shows the details of hardware developed.

IV. CONCLUSION

This paper reviews the use of multi sensor data fusion in condition monitoring of an induction machine. Multi sensor data fusion is used in condition monitoring to improve the accuracy of the diagnosis, to detect multiple faults.

REFERENCES

[1] Mo-yuen Chow et.al, "A neural network approach to real time condition monitoring of induction motors," *IEEE Transactions On Industrial Electronics*, vol. 38, no. 6, Dec 1991, pp. 448 – 453.

[2] Shaotang Chen et.al, "A new approach to motor condition monitoring in induction motor drives," *IEEE Transactions on Industry Applications*, vol. 30, no. 4, July-Aug 1994, pp. 905 – 911.

[3] Olav Vaag Thorsen et.al, "A survey of faults on induction

motors in offshore oil industry, petrochemical industry, gas terminals and oil refineries," *IEEE Transactions On Industrial Electronics*, vol. 31, no. 5, Sep/Oct 1995, pp. 1186 – 1196.

[4] Siman Atlug etal, "Fuzzy inference systems implemented on neural architecture for motor fault diagnosis," *IEEE Transactions On Industrial Electronics*, vol. 46, no.6, Dec 1996, pp. 1069-1079

[5] Lane M D Owsley et.al, "Self-organizing feature maps and hidden markov models for machine tool monitoring," *IEEE Transactions on Signal Processing*, vol. 45, no. 11, Nov 1997, pp. 2787 – 2798

[6] Md. El Hachemi Benbouzid, "A review of induction motor induction mot or signature analysis as a medium for fault detection," *IEEE Transactions in Industrial Electronics*, vol. 47, no. 5, Oct 2000, pp. 984 – 993

[7] Zhongming Y, Bin W (2001). "Online rotor bar breakage detection of three phase induction motors by wavelet packet decomposition and artificial neural network," *IEEE 32nd Annu. Conf. Power Electr. Specialists*, 4, pp. 2209–2216.

[8] X Z Gao et.al, "Soft computing methods in motor fault diagnosis," *Applied soft computing*, 1, 2001, pp. 73 – 81

[9] G K Singh, "Multi-phase induction machine drive research," *Electric Power Systems Research*, 61, 2002, pp. 139 – 147

[10] Fredrick C Trutt et.al, "Online condition monitoring of induction motors," IEEE Transactions On Industrial Electronics, vol. 38, no. 6, Dec 2002, pp. 1627 – 1632

[11] G K Singh et.al, "Induction machine drive condition monitoring and diagnostic research- - a survey," *Electric Power Systems Research,* 64, 2003, pp. 145 – 158

[12] G K Singh et.al, "Experimental investigations on induction machine condition monitoring and fault diagnosis using digital signal processing techniques," *Electric Power Systems Research*, 65, 2003, pp. 197 – 221

[13] B Samanta et.al, "Artificial neural networks and support vector machines with genetic algorithm for bearing fault detection," Engineering Applications of Artificial Intelligence, 16, 2003, pp. 657 – 665

[14] Romero-Troncoso Rene de Jesus et.al, "FPGA based on-line tool breakage detection system for CNC milling machine," *Mechatronics*, 14, 2004, pp. 439 – 454

[15] Hamid H Nasiri AR, Nasiri, "Detection and isolation of mixed eccentricity in three phase induction motor via wavelet packet decomposition," *5th Asian Control Conference*, 2004, pp. 1371 - 1376

[16] Woei Wan Tan et.al, "A generic neurofuzzy model based approach for detecting faults in induction motors," *IEEE Transactions On Industrial Electronics*, vol. 52, no. 5, Oct 2005, pp. 1420 – 1427

[17] Subhasis Nandi et.al, "Condition monitoring and fault diagnosis of electrical motors- a review," *IEEE Transactions on Energy Conservation*, vol. 20, no. 4, Dec 2005, pp. 719 – 729

[18] Sang.Bin Lee et.al, " An online technique for monitoring the insulation condition of AC machine stator windings," *IEEE* Transactions on energy conservation, vol.20, no.4, Dec 2005, pp.737-745.

[19] Antonino JA, Riera M, Roger Folch J, Molina MP, "Validation of a new method for the diagnosis of rotor bar failures for via wavelet transformation in industrial induction machine 5th *IEEE Int. Symp. Diagnost. Electric Machines*, Power Electron.Drives, SDEMPED. DEMPED, 2005, pp. 1–6.

[20] Antonino-Daviu JA, Riera-Guasp MFJR, Palomares MPM "Validation of a new method for the diagnosis of rotor bar failures via wavelet transform in industrial induction machines," *IEEE Trans. Ind. Appl.*, 42(4), 2006, pp. 990– 996

[21] A R Mohanty et.al, "Monitoring gear vibrations through motor current signature analysis and wavelet transform," *Mechanical systems and signal processing*, 20, 2006, pp. 158-187

[22] Md. H Kahaei et.al, "Detection of bearing faults using Harr wavelets," *IEICE Transactions*, vol. 89, no. 3, Mar 2006, pp. 757 – 763

[23] Fulufhelo V Nelwamondo et.al, "Early classification of bearing faults using HMMs, GMMs, MFCC and fractals," *Int. J. Innovative computing, information and control*, vol. 2, no. 6, Dec 2006, pp. 1281 – 1299

[24] Tin Han et.al, "Fault diagnosis system of induction motors based on neural network and genetic algorithm using stator current signal," *Hindwai Publishing Corporation*, Int. J. Rotating Machinery, vol. 2006, 2006, pp. 1-13

[25] U Natarajan et.al, "A decision fusion algorithm for tool condition monitoring in drilling using hidden markov models," *Indian Journal of Engineering and Material Sciences*, vol. 13, April 2006, pp. 103 – 109

[26] Antonino-Daviu, J. A.; Riera-Guasp, M.; Folch, J. R. & Palomares, M. P. M., "Validation of a new method for the diagnosis of rotor bar failures via wavelet transform in industrial induction machines, *IEEE Transactions on Industry Applications*, vol. 42, no. 4, 2006, pp. 990- 996

[27] Andrew K S Jardine et.al, "A review on machinery diagnostics and prognostics implementing condition based maintenance," *Mechanical systems and signal processing*, 20, 2006, pp. 1483 – 1510

[28] Ming Dong et.al, "Hidden Semi-Markov model based methodology for multi sensor equipment health diagnosis and prognosis," *European Journal of Operational Research*,17, 8, 2007, pp. 858 – 878

[29] Bo-Suk Yang et.al, "Support vector machine in machine condition monitoring and fault diagnosis," *Mechanical systems and signal processing*, 21, 2007, pp. 2560 – 2574

[30] Ashraf Saad et.al, "Evolving an artificial neural network classifier for condition monitoring of rotating mechanical system," *Applied Soft Computing*, 7, 2007, pp. 441 – 454

[31] Rangarajan M. Tallam et.al, "A survey of methods for detection of stator related faults in induction machines," *IEEE Transactions on Industry Applications,* vol. 43, no. 4, July/Aug 2007, pp. 920 – 933

[32] Z Jafar and P Javad, "Bearing fault detection using wavelet packet transform of induction motor stator current," *Tribol. Int.*, 40: 763–769, 2007

[33] Bin L, Paghda M (2008). Induction motor rotor fault diagnosis using wavelet analysis of one-cycle average power" Twenty-Third Annual *IEEE conf. Appl. Power Electron. Exposit. APEC*,

pp. 1113–1118.

[34] Alberto Bellini et.al, "Advances in diagnostic techniques for induction machines," *IEEE Transactions On Industrial Applications*, `vol.55, no.12 Dec 2008, pp.4109-4126

[35] Pratesh Jayaswal et.al, "Machine fault signature analysis," *Hindwai Publishing Corporation, Int. J. Rotating Machinery*, vol. 2008, 2008, pp. 1-10

[36] Luis Alfonso Franco-Gasca et.al, "FPGA based failure monitoring system for machining process," *Int. J. A advance Manufacturing Technology*, 40, 2009, pp. 676 – 686 .

[37] Subramaniam Manoharan et.al, "Review on efficiency improvement in squirrel cage induction motor by using DCR technology," *Journal of electrical engineering*, vol.60, no.4, 2009, pp.227-236

[38] F Jawad and O Mansour, "Different indexes for eccentricity fault diagnosis in three phase squirrel cage induction motor: a review," *Mechatronics,* 19: 2-13, 2009.

[39] Luis Miguel Contreas-Medina et.al, "FGPA based multi channel vibration analyzer for industrial application in induction motor failure detection," *IEEE Transactions on Instrumentation and Measurement*, vol.59, no,1, Jan 2010, pp.63-72

[40] Gang Niu et.al, "Development of an optimized condition based maintenance system by data fusion and reliability centered maintenance," *Reliability Engineering and System Safety*, 95, 2010, pp.786 – 796

[41] Yixiang H, Chengliang L, Xuan FZ, Yanming L, "A lean model for performance assessment of machinery using second generation wavelet packet transform and Fisher criterion." *Exp. Syst. Appl.*, 2010, pp. 3815–3822.

[42] Rene J Romero-Troncoso et.al, "FPGA based online detection of multiple faults in induction motors through information entropy and fuzzy inference," *IEEE Transactions on Industrial Electronics*, vol. 58, no. 11, Nov 2011, pp. 5263 – 5270.